

TRAVIS W. BATZEL

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PORTFOLIO

QR CODE

EDUCATION

North Central College, Naperville, IL | **B.S. Computer Engineering** — Fall 2028

Minor in Artificial Intelligence • College Scholars Honors Program

GPA: 3.90 • LEAPS Scholar • Navistar Emerging Leaders Scholarship

Relevant Coursework: Circuit Analysis • Introduction to Electrical Engineering • Data Structures • Software Development • Microprocessor Assembly Language • Classical Artificial Intelligence • Engineering Methods • Calculus II

College of DuPage, Glen Ellyn, IL | **Associate of Arts, High Honors**

RESEARCH EXPERIENCE

3-Axis Hall-Effect Magnetometer — Summer Undergraduate Research | North Central College, Summer 2026

- Developed a 3-axis magnetic-field measurement system using Hall probes, custom analog signal conditioning, a 16-bit ADS8688 ADC, and Arduino Uno R4 Minima.
- Implemented SPI-based data acquisition and firmware for X/Y/Z measurements; contributed to KiCad PCB design, grounding/shielding, power distribution, enclosure design, and system integration.
- Characterized system performance and presented the completed research at the Summer Research Symposium.

SELECTED PROJECTS

Swoolly — iOS Diet & Workout Tracker | Independent Project

- Designed and developed a native iOS application using SwiftUI and SwiftData for nutrition, weight, activity, and workout tracking.
- Implemented persistent food and exercise logs, macro tracking, historical data, meal presets, progress visualization, and data backup/import-export.

Audio Guestbook — Embedded Audio System | Independent Project

- Designed and built an ESP32-based telephone guestbook integrating a WM8960 audio codec, microSD storage, I²S audio, and physical handset controls.
- Implemented automated greeting and tone playback, audio recording, and file storage controlled by handset state.

Three Card Poker AI Agent | Academic Project

- Developed a Python Three Card Poker agent using hand-strength analysis and expected-value calculations to make autonomous gameplay decisions.
- Implemented persistent gameplay history and opponent modeling to adapt decisions based on observed betting behavior.

TECHNICAL SKILLS

Embedded/Electronics: Arduino, ESP32, SPI, I²S, ADC integration, circuit analysis, schematic reading, soldering

Programming: Swift/SwiftUI, C++, Python; familiarity with Java and assembly

Engineering: SolidWorks, KiCad PCB Design, 3D printing, electronics prototyping, signal conditioning

Data: SQL, Power BI, SwiftData

PROFESSIONAL EXPERIENCE

Top Shelf Expeditors, Elmhurst, IL — Domestic Freight Broker | 2022–2024

- Managed high-volume, time-sensitive operations requiring independent troubleshooting and coordination.

Clutch Global Logistics, Oak Brook, IL — North American Trucking Manager | 2018–2022

- Developed SQL-based reporting systems and Power BI dashboards; led implementation of operational tracking tools.

MILITARY SERVICE

United States Marine Corps | 2010–2016 • Honorable Discharge